Course 2 Lab: Summarize Private Docs with RAG & LangChain

Course 2 Lab: Summarize Private Documents Using RAG, LangChain, and LLMs

This is your distilled, "No-Scrolling" cheat sheet for the Course 2 RAG Lab.

[SKIP] The SKIP Pile (Boilerplate / Busywork)

- Setup & Installs: pip install ibm-watsonx-ai, wget.download(). Basic environment setup.
- Downloading companyPolicies.txt: Simple text download.
- Interactive while True: Loop: Terminal-style input loop.

[SKIM] The SKIM Pile (Read & Understand Concept)

- PromptTemplate Guardrails: Adding "If you don't know the answer, just say that you don't know" to RetrievalQA.
- Why? Prevents the LLM from hallucinating answers when the retrieved context doesn't contain the answer.
- ConversationalRetrievalChain + ConversationBufferMemory:
- Why skim? This is legacy LangChain syntax for memory in RAG chains. In modern 2026 production (Course 7), memory and conversational state in RAG are handled using LangGraph state channels.

[MUST KNOW] The STRUGGLE Pile (Core Interview Knowledge)

Focus your effort on these concepts and exercises.

1. Document Indexing Flow (The 3 Steps)

1. Load: TextLoader(filename).load() -> Converts raw .txt into LangChain Document objects.
2. Split: CharacterTextSplitter(chunk_size=1000, chunk_overlap=0) -> Chops documents into smaller chunks.
3. Embed & Store: HuggingFaceEmbeddings() -> Turns text into vectors. Chroma.from_documents(texts, embeddings) -> Stores vectors and text in Chroma vector database.

2. Exercise 2: Source Citation (return_source_documents=True)

- Interview Question: "How do you show users the exact paragraph/source document where your RAG system found its answer?"
- Answer: In LangChain, set return_source_documents=True when building your QA chain. When you call qa.invoke(query), the returned dictionary contains a 'source_documents' list containing the exact Document chunks retrieved from ChromaDB!

python

Key syntax to remember:

```
results = qa.invoke(query)
first_source = results['source_documents'][0]
print(first_source.page_content)  # Raw text chunk
print(first_source.metadata)      # Source file / page info
```

3. Exercise 3: Model Swapping & Evaluation

- Interview Concept: Swapping model IDs (ibm/granite-3-8b-instruct vs mistralai/mistral-small-3-1-24b-instruct-2503) to evaluate model performance, context window limits, and answer quality on the exact same retrieved RAG context.

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RAG Execution Flow Summary

Raw File (.txt) -> TextLoader -> CharacterTextSplitter -> HuggingFaceEmbeddings -> Chroma VectorStore -> RetrievalQA -> Grounded Answer + Source Citations